

Configuration

The sd.conf file, the two ways to read a setting, and every parameter SD accepts.

SD reads one configuration file at start-up. It sizes the shared memory segment, sets the limits every session inherits, and names the directories SD writes to.

This document is separate so that it can be withheld. It links to nothing outside the administrator set. Where a page in another set is worth naming, it is named in words.

SD folds case, so a command may be typed in either case. Commands are shown here in lower case.

Every listing on this page was produced by running it, on SD Core for Windows W1.0-0, from an unelevated session in an administrator account.

The file

C:\ProgramData\SD\sd.conf

The server and the client both read the `SD_CONFIG` environment variable first and fall back to that path. The file is installed only if it does not already exist and is marked never to uninstall, so edits to it survive an upgrade and survive removal of the product.

It is plain text in one section:

```
[sd]
SDSYS=C:\ProgramData\SD\sdsys
GRPSIZE=2
NUMUSERS=20
```

Lines beginning `#` are comments. The shipped file is heavily commented and those comments record why each value was chosen. Read them before changing anything.

A NAME SD DOES NOT RECOGNISE STOPS IT STARTING

An unrecognised parameter is not ignored. The parser abandons the file with `Unrecognised configuration parameter`, and SD does not start.

That is why obsolete names are still parsed rather than deleted. `CREATUSR` has done nothing since 14 August 2026, but a configuration file copied from a Linux installation still carries it, and removing the branch would turn a tidy-up into a failure to start.

Range failures behave the same way. `GRPSIZE`, `INTPREC`, `LPTRHIGH`, `LPTRWIDE`, `MAXCALL`, `RECCACHE`, `SORTMRG` and `MAXIDLEN` are bounds-checked once the file has been read, and a value outside its range stops start-up with a message naming the parameter.

Reading the settings

The `config` verb reports what is in force:

```
:config
Virtual Machine Version Number W1.0-0
APILOGIN  1
APIPORT   4243
CMDSTACK  99
DEADLOCK  0
DUMPDIR   C:\ProgramData\SD\sdsys\dumps
ERRLOG    50 kb
...
YEARBASE  1930
```

SD accepts **52** parameters and the verb prints **43** of them. Nine are accepted in the file and never displayed: `CODEPAGE`, `CREATUSR`, `DEBUG`, `FDS`, `FIXUSERS`, `NETDIRS`, `PORTMAP`, `SDSYS` and `TXCHAR`.

`NETDIRS` is the one to know about. It decides what an API session may reach outside its own account, and the verb will not tell you what it is set to. Read it from the file.

The `config()` function reads one parameter from a program:

```
group.size = config('GRPSIZE')
```

49 parameters are readable this way. The three that are not are `CREATUSR`, which stores nothing, `SDSYS`, and `TXCHAR`. The name is truncated to eight characters before it is matched; no parameter name is longer than eight, so that only shows up if you pass something which is not a parameter.

Two further forms exist. `config_lptr` reports the settings of the default printer, and `config_gpl` and `config_contrib` display the licence and the list of contributors.

Changing a parameter

Editing `sd.conf` and restarting SD is the durable route, and for most parameters it is the only one.

28 parameters can also be changed for the current session:

```
config sortmem 8192
```

That writes to the session's own copy of the settings, taken from shared memory when the session started. It does not reach `sd.conf`, it does not affect any other session, and it is gone when the session ends.

The 28 are `CODEPAGE`, `DUMPDIR`, `EXCLREM`, `FILERULE`, `FLTDIFF`, `FSYNC`, `GDI`, `GRPSIZE`, `INTPREC`, `LPTRHIGH`, `LPTRWIDE`, `MAXCALL`, `MUSTLOCK`, `OBJECTS`, `OBJMEM`, `RECCACHE`, `RINGWAIT`, `SAFEDIR`, `SDCLIENT`, `SH`, `SH1`, `SORTMEM`, `SORTMRG`, `SORTWORK`, `SPOOLER`, `TEMPDIR`, `TERMINFO` and `YEARBASE`.

Everything else takes effect only when SD is next started. That includes every limit which sizes the shared memory segment and every setting the API listener reads.

Sessions and limits

PARAMETER	DEFAULT	EFFECT
<code>NUMUSERS</code>	20	Maximum concurrent sessions. Sizes the user table in shared memory
<code>NUMFILES</code>	80	Maximum open files across all sessions
<code>NUMLOCKS</code>	100	Maximum record locks across all sessions
<code>MAXCALL</code>	10000	Maximum subroutine call depth. Range 10 to 1000000
<code>CMDSTACK</code>	99	Depth of the command stack
<code>FDS</code>	unset	Limit on file descriptors. No limit when unset
<code>FIXUSERS</code>	unset	<code>base,range</code> — user numbers reserved for sessions that ask for a specific number
<code>PORTMAP</code>	unset	<code>base_port,base_user,range</code> — gives a session a fixed user number derived from the port its connection arrived on. Refused if the range overlaps <code>FIXUSERS</code>

Files and locking

PARAMETER	DEFAULT	EFFECT
GRPSIZE	2	Default group size for a new dynamic file, in 1 KB units. Read by <code>create.file</code> and <code>configure.file</code> when no group size is given
MAXIDLEN	63	Maximum record id length. 63 is also the lower bound
MUSTLOCK	0	When 1, a <code>write</code> or <code>delete</code> requires the record to be locked first
DEADLOCK	0	When 1, SD traps deadlocks
SAFEDIR	0	When 1, directory files are updated by write-and-rename rather than in place
RECCACHE	0	Records cached per file. Range 0 to 32
FSYNC	0	Bit flags controlling when SD forces data to disk
FILERULE	0	Bit flags deciding which special VOC file references are honoured. Bit 4 allows a <code>PATH:</code> reference to name a pathname directly

Directories

PARAMETER	DEFAULT ON A NEW INSTALL	EFFECT
SDSYS	<code>C:\ProgramData\SD\sdsys</code>	The system account. SD does not start if the global catalogue is not found beneath it
USRDIR	<code>C:\ProgramData\SD\user_accounts</code>	Where <code>create.account</code> puts a user account
GRPDIR	<code>C:\ProgramData\SD\group_accounts</code>	Where <code>create.account</code> puts a group account
DUMPDIR	<code>C:\ProgramData\SD\sdsys\dumps</code>	Where process dumps are written
TEMPDIR	<code>/cygdrive/c/WINDOWS/TEMP</code>	Temporary files
SORTWORK	<code>/cygdrive/c/WINDOWS/TEMP</code>	Work files for a sort that does not fit in memory
JNLDIR	empty	Journal directory
TERMINFO	empty	An additional terminfo directory. The shipped definitions are found without it

`DUMPDIR` is set rather than left empty on purpose. A blank value falls back to the system directory, which SD users can write to, and a process dump carries the whole variable state of the session that wrote it. The installer makes the dump directory write-only to SD users, so a dump can be added and nobody else's can be listed or read.

`TEMPDIR` and `SORTWORK` are reported in POSIX form because that is how the server's runtime addresses them.

`C:\WINDOWS\TEMP` and `/cygdrive/c/WINDOWS/TEMP` are the same directory.

The API

PARAMETER	DEFAULT	EFFECT
<code>APIPORT</code>	4243	The port the API listens on. If the line is absent no socket is created at all, which is how the API is turned off
<code>APILOGIN</code>	1	Whether the API requires authentication. <code>0</code> is the weaker setting, not the safer one
<code>NETDIRS</code>	unset	Directories outside its own account an API session may open, separated by semicolons because a Windows path contains a colon
<code>SDCLIENT</code>	0	Restricts what an API session may do. Non-zero disables file access outright; <code>2</code> additionally refuses any subroutine not compiled as callable from a client

Unset is the strict value for `NETDIRS`. With nothing there, an API session can open files in the account it is standing in and nothing else. It never grants the credential store, the global catalogue, `os.users` or the account register, and naming those has no effect. A directory listed here is reachable by every API session in every account, so it is a decision about the machine rather than about one account.

`SDCLIENT` is not reported by the `config` verb and has no entry in the shipped `sd.conf`. It defaults to 0, which permits everything.

Printing

PARAMETER	DEFAULT	EFFECT
<code>LPTRWIDE</code>	80	Default page width for the printer. Range 10 to 1000
<code>LPTRHIGH</code>	66	Default page depth for the printer. Range 10 to 32767
<code>SPOOLER</code>	empty	The spooler to hand print jobs to
<code>GDI</code>	0	Selects the Windows printing interface used by default

Sorting

PARAMETER	DEFAULT	EFFECT
<code>SORTMEM</code>	4096 kb	Above this much data a sort works on disk instead of in memory
<code>SORTMRG</code>	4	Files merged at once in a disk sort. Range 2 to 10

Numbers and dates

PARAMETER	DEFAULT	EFFECT
<code>INTPREC</code>	13	Digits of precision in integer arithmetic. Range 0 to 14
<code>FLTDIFF</code>	0.00000000002910	Two floating point numbers closer together than this compare equal
<code>YEARBASE</code>	1930	The century a two-digit year is read into

Diagnostics

PARAMETER	DEFAULT	EFFECT
<code>ERRLOG</code>	50 kb	Size of the error log. A non-zero value below 10 KB is raised to 10 KB, and the oldest entries are discarded when it fills
<code>PDUMP</code>	0	Bit flags controlling when a process dump is written
<code>DEBUG</code>	unset	Bit flags enabling debugging features
<code>STARTUP</code>	empty	A command run when SD starts
<code>JNLMODE</code>	0	Journalling mode
<code>OBJECTS</code>	0 (no limit)	Compiled programs held in memory at once
<code>OBJMEM</code>	0 (no limit)	Memory those programs may occupy, in KB

The shell

PARAMETER	DEFAULT	EFFECT
SH	PowerShell, interactive	The shell a bare sh starts
SH1	PowerShell, non-interactive	The shell sh command uses

Both are full paths on a real install:

```
SH      C:/Windows/System32/WindowsPowerShell/v1.0/powershell.exe -NoProfile -NoLogo
SH1     C:/Windows/System32/WindowsPowerShell/v1.0/powershell.exe -NoProfile -NonInteractive
```

The difference between them matters. SH1 carries -NonInteractive and SH does not, so a bare sh in a phantom or a scheduled job hands control to a shell with nobody at the keyboard. Operating system access has its own page in this set.

Parameters that are accepted and do nothing

These are parsed so that an existing sd.conf still loads. None of them changes SD's behaviour, and none should be offered to a site as a control.

PARAMETER	WHY IT IS INERT
NETFILES	SDNet was removed from this port. netfiles.c is deleted, a server;file VOC reference is refused, and the request that reports open SDNet connections returns an empty list. The value is still stored and still reported, and setting it opens nothing
CREATUSR	SD has accounts rather than accounts and users, so create.account always creates the operating system account and there is nothing to opt in to. The value is discarded as it is read
CODEPAGE	Stored, readable and settable. Nothing acts on it
EXCLREM	Stored, readable and settable. It described exclusive access to a remote file, and there are no remote files
RINGWAIT	Stored, readable and settable. Nothing acts on it
TXCHAR	Stored, and not readable — there is no config('TXCHAR')

`FILERULE` is **not** in this list, and is easy to mistake for it. Its remote bits died with SDNet, but bit 4 is live and decides whether a `PATH:` VOC reference may name a pathname directly.

There is no licence parameter and no licence verb. SD Core for Windows is GPL v3 and is not licensed per site or per user; `config_gpl` displays the licence text.